

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
6 using System.Text.Json;
7 using System.Text.Json.Serialization;
8 using CustomProjectRPG.Interfaces;
9
10 namespace CustomProjectRPG
11 {
12
13
14
15
16     public class UIManager : ISerializeJSON<UIManager>, IUIManager
17     {
18         private string _lastCorruptionDisplay;
19         private Dictionary<string, bool> _lastQuestDisplay; // questId -> isComplete
20         private string _lastDialogueDisplay;
21         private bool _isVisible;
22
23         public event Action<string> UIRefreshed; // Event for UI refresh notifications
24
25
26         // Initializes a new UIManager with default display states.
27
28         public UIManager()
29         {
30             _lastCorruptionDisplay = "No corruption data.";
31             _lastQuestDisplay = new Dictionary<string, bool>();
32             _lastDialogueDisplay = "No dialogue available.";
33             _isVisible = true;
34         }
35
36
37         // Updates the corruption display based on entity state.
38
39         public void UpdateCorruptionDisplay(string entityId, CustomProjectRPG.Entities.Enum.Enum.CorruptionState state)
40         {
41             _lastCorruptionDisplay = $"Entity {entityId} Corruption State: {state}";
42             Render();
43             UIRefreshed?.Invoke(_lastCorruptionDisplay);
44         }
45     }
```

```
46
47     // Updates the quest display based on quest status.
48
49     public void UpdateQuestDisplay(string questId, bool isComplete)
50     {
51         _lastQuestDisplay[questId] = isComplete;
52         _lastQuestDisplay.TryGetValue(questId, out bool currentState);
53         _lastQuestDisplay[questId] = currentState; // Ensure entry exists
54         Render();
55         UIRefreshed?.Invoke($"Quest {questId} Status: {(isComplete ? "Complete" : "In Progress")}");
56     }
57
58
59     // Updates the dialogue display with new text.
60
61     public void UpdateDialogueDisplay(string dialogue)
62     {
63         _lastDialogueDisplay = dialogue;
64         Render();
65         UIRefreshed?.Invoke(_lastDialogueDisplay);
66     }
67
68
69     // Toggles UI visibility.
70
71     public void ToggleVisibility()
72     {
73         _isVisible = !_isVisible;
74         Render();
75     }
76
77
78     // Renders the current UI state to the console (placeholder for graphical UI).
79
80     private void Render()
81     {
82         if (!_isVisible) return;
83         Console.Clear();
84         Console.WriteLine("=== Game UI ===");
85         Console.WriteLine(_lastCorruptionDisplay);
86         Console.WriteLine("Quests:");
87         foreach (var quest in _lastQuestDisplay)
88         {
89             Console.WriteLine($"    {quest.Key}: {(quest.Value ? "Complete" : "In Progress")}");
90         }
```

```
91         Console.WriteLine("Dialogue:");
92         Console.WriteLine($"    {_lastDialogueDisplay}");
93         Console.WriteLine("=====");
94     }
95
96
97
98
99     public string Serialize()
100     {
101         var data = new UIData
102         {
103             LastCorruptionDisplay = _lastCorruptionDisplay,
104             LastQuestDisplay = _lastQuestDisplay,
105             LastDialogueDisplay = _lastDialogueDisplay,
106             IsVisible = _isVisible
107         };
108         return JsonSerializer.Serialize(data);
109     }
110
111
112
113
114     public void Deserialize(string json)
115     {
116         var data = JsonSerializer.Deserialize<UIData>(json)
117             ?? throw new JsonException("Invalid JSON format for UI state.");
118         _lastCorruptionDisplay = data.LastCorruptionDisplay;
119         _lastQuestDisplay = data.LastQuestDisplay ?? new Dictionary<string, bool>();
120         _lastDialogueDisplay = data.LastDialogueDisplay;
121         _isVisible = data.IsVisible;
122         Render();
123     }
124
125     private class UIData
126     {
127         [JsonPropertyName("lastCorruptionDisplay")]
128         public string LastCorruptionDisplay { get; set; }
129
130         [JsonPropertyName("lastQuestDisplay")]
131         public Dictionary<string, bool> LastQuestDisplay { get; set; }
132
133         [JsonPropertyName("lastDialogueDisplay")]
134         public string LastDialogueDisplay { get; set; }
135
136         [JsonPropertyName("isVisible")]
137         public bool IsVisible { get; set; }
```

```
138         }  
139     }  
140 }  
141
```